Harsh Solanki

Gen AI Engineer

+91-9315605618 Faridabad, Haryana harsh.solanki1520@gmail.com linkedin.com/in/harshsolanki
github.com/Firehawk39 HuggingFace: Anachronox39

Summary

Generative AI Engineer with proven experience building local-first, privacy-first AI systems from fine-tuning LLMs and deploying RAG pipelines to orchestrating multi-agent workflows on self-hosted GPU infrastructure. Passionate about applying AI to real-world problems. Curious, experimental, and constantly building.

Skills

Programming Languages: Python, C, HTML
AI/ML Frameworks: PyTorch, TensorFlow, LangGraph, LangChain, Unsloth Studio, Scikit-learn, NumPy, Pandas
Generative AI & LLM: LLM Fine-tuning (QLoRA), RAG, Prompt Engineering, Hugging Face, Transformers, Diffusion Models, Local LLM Deployment, NLP
Agentic AI & Automation: OpenClaw, MCP, n8n, Google Antigravity, Claude Code, Lovable, Ollama, ComfyUI, Wan2GP
Vector & Database: Pinecone, FAISS, Chroma, PostgreSQL, MongoDB, Supabase
Deployment & Cloud: AWS, GCP, Microsoft Azure, Hugging Face Spaces, Docker, Git, Linux, HomeLab
Soft Skills: Analytical Thinking, Problem-Solving, Team Collaboration, Effective Communication

Experience

Generative AI Intern **Jul 2025 – Sep 2025**
Manav Rachna Centre for Health Innovations

- Architected and managed end-to-end local GPU-accelerated GenAI workflows using Wan2.1, FLUX, and LTX-2, **improving content iteration speed by 20%** while ensuring full data privacy.
- Built **2 automation pipelines** for short-form health awareness video production, delivering **4 videos** on stroke, seizure, and other emergencies – bridging Generative AI with healthcare communication.
- Produced a product marketing video for the AnemoScan mobile app; collaborated cross-functionally with healthcare and design teams to align AI output with campaign goals.

IT Intern **Jul 2024 – Aug 2024**
NTPC Ltd.

- Gained hands-on exposure to enterprise-scale IT infrastructure, supporting **3 systems** and resolving **2 tickets** for a team of **5 employees** within one of India's largest public sector organizations.
- Created **4 Standard Operating Procedures (SOPs)** for recurring hardware issues, improving resolution clarity for internal IT staff.

Projects

PersonaFlow – AI Wellness Companion | *RAG, Supabase, n8n, Ollama, Vercel* | github.com/Firehawk39 **Dec 2025**

- Built a full-stack platform featuring an empathetic AI companion powered by hermes3:8b-llama3.1, a model for emotionally intelligent, supportive dialogue. Implemented long-term memory using Supabase to store journals, habits, and reflections – enabling the AI to develop a persistent "second self" that understands the user more deeply over time.
- Built advanced AI pipelines using RAG and multi-tool agents to deliver context-aware, emotionally aligned guidance that becomes more accurate and personalized with every interaction.
- Engineered a privacy-first, 100% local AI system ensuring complete data ownership while supporting real-time semantic search and multi-step reasoning workflows through n8n automation.

NutriBud – AI Nutrition Assistant | *LLAMA 3.2 Vision, Unsloth, n8n* | HuggingFace **Sep 2025**

- Built an end-to-end Telegram Bot for automated meal photo analysis to estimate calories, protein, fat, and carbs – entirely using self-hosted n8n and Telegram Bot API.
- Significantly enhanced model accuracy by **50%** for diverse cuisine by fine-tuning LLAMA 3.2 Vision 11B on the Indian Food Images Dataset, leveraging Unsloth/QLoRA for memory-optimized training on 12 GB VRAM.

– Open-sourced the fine-tuned model on Hugging Face that substantially outperformed general VLMs on region-specific food classification accuracy.

Self-Hosted AI Infrastructure Rig | *HiveOS, Salad, Ubuntu, Vast.ai* **Dec 2024**

– Built and managed a **4x NVIDIA RTX GPU** HomeLab rig with 24/7 uptime for self-hosting AI/ML and compute monetization via Vast.ai.

– Optimized thermal management and power configuration to **reduce energy consumption by 15%** while maintaining peak GPU throughput.

Awards & Certifications

- **Azure Data Scientist Associate (DP-100) – Microsoft (2025):**
- **Azure Data Engineer Associate (DP-203) – Microsoft (2025):**
- **Azure AI Fundamentals (AI-900) – Microsoft (2024)**
- **Azure Fundamentals (AZ-900) – Microsoft (2024)**
- **Microsoft Security, Compliance & Identity (SC-900) – Microsoft (2024)**
- **Machine Learning using Python – Infosys Springboard (2024)**
- **Python for Data Science – Infosys Springboard (2024)**

Education

B.Tech – Computer Science (AI/ML Specialization) **Graduating: Jul 2026**

Manav Rachna International Institute of Research and Studies *CGPA: 7.0 / 10*